A unified viscous-spring artificial boundary for 3-D static and dynamic applications

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Abstract A method to develop unified artificial boundaries for problems coupling static effect and dynamic effect is proposed. Based on the dynamic viscous-spring artificial boundary and the fundamental solution of static problems in elastic half space, a unified viscous-spring artificial boundary for 3-D static and dynamic applications is established.

Keywords: artificial boundary, half space, open system, soil-structure interaction.

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The soil-structure system in civil engineering is an open system, in which the simulation of the half space is necessary for static problems as well as dynamic problems. Thus, it is crucial to set artificial boundary reasonably for soil-structure interaction problems. In general, the artificial boundary can be divided into static boundary and dynamic boundary. The static artificial boundary is a conventional boundary with a long history, such as the fixed boundary and the roller boundary. Most of the dynamic artificial boundaries used widely at present belong to the local boundary based on the one-way wave theory, such as the viscous boundary[1], the viscous-spring boundary[2−5] and the Liao's boundary[6]. The viscous boundary with the clear definition is simple and widely applied[1,7]. But it is of low accuracy and unstable in low frequency[3]. Oppositely, the viscous-spring boundary with good stability and high accuracy can constrain the zero-frequency component and simulate the elasticity recovery capacity of the far field media out of the artificial boundary[2,4]. Currently, for engineering problems coupling static and dynamic effect, static analysis and dynamic analysis are completed respectively based on the static boundary and dynamic boundary, and a total solution is achieved by superposition. However, the superposition principle is limited to linear elasticity and small deformation, so it cannot be applied to the problems with inelasticity or large deformation, such as the seismic response analysis of the large arch-dam, the high structure-soil system under the consideration of P-Δ effect, and influence analysis on the dynamic performance of underground structures by the static loads of the soil-structure system. Therefore, it is necessary to develop a unified artificial boundary applicable to

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both static and dynamic problems.

Generally, the static and dynamic artificial boundaries are developed respectively on the basis of the statics and the dynamics. It is difficult to establish a unified boundary from both basic theories. A feasible method is to develop a dynamic (static) artificial boundary for dynamic (static) problems based on the dynamic (static) theory firstly, and then to modify it adaptively to static (dynamic) problems. With this idea, a unified viscous-spring artificial boundary for 3-D static and dynamic applications is presented in this paper.

1 Viscous-spring dynamic artificial boundaries and their robustness analysis

In order to overcome the disadvantages of the viscous boundary, such as inaccuracy and instability in low frequency, Deeks[2] and Liu[3,4] proposed the 2-D viscous-spring dynamic boundary on the assumption that the scattered wave is the 2-D cylindrical wave. And then, Liu[5] proposed the 3-D viscous-spring dynamic artificial boundary on the presumption that the scattered wave was the 3-D spherical wave. Both the above viscous-spring dynamic boundaries can be implemented by distributing the spring/dashpot system, in which the spring and the dashpot are placed in parallel. The spring constant K_b and the damping coefficient C_b can be expressed as follows:

$$K_b = \alpha \frac{G}{R}, \tag{1}$$

$$C_b = \rho c, \tag{2}$$

where ρ is the mass density; G is the shear modulus respectively; R is the distance between the wave source and the artificial boundary; c is the wave velocity, P-wave velocity c_p and S-wave velocity c_s are adopted respectively in the normal and tangent direction; parameter α is set as Table 1 according to the type and direction of the artificial boundary, and in the following text, α_n and α_t denote the normal and tangent parameter α respectively. Apparently, when $\alpha = 0$ or R tends to be infinite, the stiffness K_b equals to zero. Thus, the viscous-spring artificial boundary degenerates to the viscous boundary.

Table 1 Parameter α in viscous-spring dynamic boundaries

Type	Direction	α
2-D boundary	normal in plane	2.0
	tangent in plane	1.5
	tangent out of plane	0.5
3-D boundary	normal	4.0
	tangent	2.0

It is easy to apply the viscous-spring dynamic boundaries in the finite element method by placing the parallel spring element and damper element in the normal and tangent direction on every finite element node of the artificial boundary[5,8]. Fig. 1 shows the

method to implement the 3-D viscous-spring dynamic boundary, in which x and y are the tangent directions in the artificial boundary plane, and z is the normal direction. The stiffness K of the spring element and the damping coefficient C of the dashpot element can be defined easily according to eqs. (1) and (2),

$$K = \alpha \frac{G}{R} \sum A_i, \tag{3}$$

$$C = \rho c \sum A_i, \tag{4}$$

where $\sum A_i$ denotes the area around the artificial boundary node, that is, the part surrounded by broken line in Fig. 1.

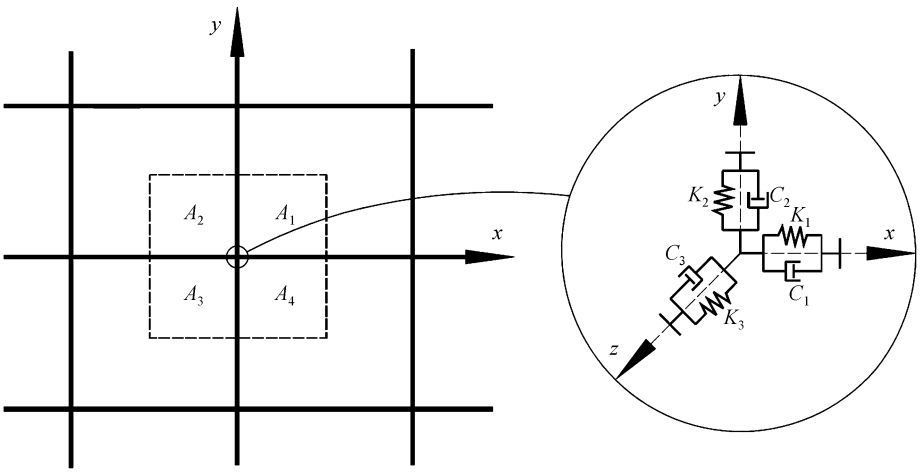

Fig. 1. Implementation of 3-D viscous-spring dynamic boundary.

To see the robustness of the 3-D viscous-spring dynamic boundary, the inner source problem in the 3-D infinite space is analyzed. Fig. 2(a) shows the analysis model with $\rho = 1$, $c_s = 4$ and $\nu = 0.25$. The force $P(t)$ is set as the approximation of Dirac's δ function by finite differentiation, that is,

$$\delta(\tau) = 16\left[G_4(\tau) - 4G_4\left(\tau - \frac{1}{4}\right) + 6G_4\left(\tau - \frac{1}{2}\right) - 4G_4\left(\tau - \frac{3}{4}\right) + G_4(\tau - 1)\right], \tag{5}$$

where $\tau = t/T$ is the dimensionless time; T is the duration ($T = 1.0$ s); $G_4(t) = \tau^3 H(\tau)$, $H(\tau)$ is the Heaviside function. Although it may be more accurate to establish the spherical boundary with the center on the loading point, the plane artificial boundary is used in this numerical example as well as in the practical engineering. The computational region is taken $-0.5 \leqslant x, y, z \leqslant 0.5$ with the finite element size taken $\Delta x = \Delta y = \Delta z = 0.05$. For simplification, R on every boundary plane is taken the same value and the vertical distance between the loading point and the boundary plane is adopted in this example, i.e.

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R = 0.5. The robustness of the 3-D viscous-spring dynamic artificial boundary is investigated by changing the stiffness K_b given in eq. (1). And parameter $\alpha_n = 2\alpha_t$ is taken 4.0, 2.0, 4/3, 1.0 and 0.0 respectively, among which 4.0 and 0.0 correspond to the 3-D viscous-spring dynamic boundary and the viscous boundary separately. Fig. 2(b) shows the displacement time history in the z direction of the observation point (0.2, 0.2, 0.2), from which it is known that when the stiffness is between 0 and the value derived in eq. (1), the accuracy of the viscous-spring dynamic boundary is higher than that of the viscous boundary. When α_n varies between 1.0 and 2.0, the viscous-spring boundary is always in good accuracy. The above results show that the viscous-spring dynamic boundary has good robustness, which provides guarantee to develop a unified boundary by modifying the stiffness K_b. When R is taken 0.5, 1.0, 1.5, 2.0, and ∞, the same results can be obtained, which demonstrates that R in the viscous-spring dynamic boundary can change in

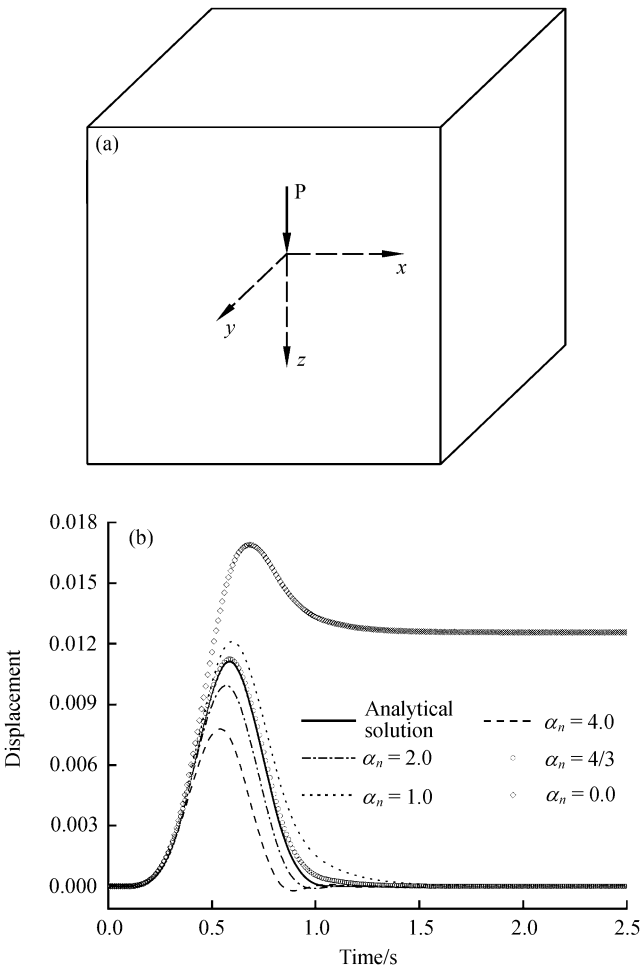

Fig. 2. Inner source problem. (a) Analysis model; (b) displacement of observation point in z-direction.

a wide range; that is, the value of R also has good robustness.

The viscous-spring dynamic artificial boundaries have clear definition, simple formulation, high accuracy, and good stability, so that they have been applied successfully to the soil-structure dynamic interaction problems[8−10]. However, these boundaries are developed on the basis of the wave theory in the homogenous elastic infinite space. Although the spring ensures the isolated system stable, the direct use in static problems may result in large error. In addition, there exists the surface wave such as the Rayleigh wave in half space besides P-wave and S-wave, which determines the difference of the elastic wave field between half space and infinite space, especially in the vicinity of the surface. Therefore, in order to simulate the elastic half space model in engineering more reasonably, it is necessary to modify the 3-D viscous-spring dynamic boundary[11].

2 A unified viscous-spring artificial boundary in 3-D static and dynamic problems

The determination of the dynamic artificial boundary is a problem of low frequency[12]. Refs. [11, 13] have pointed out that the accuracy of a viscous-spring artificial boundary is governed by its static behavior. According to this rule, if the viscous-spring dynamic boundary can be modified adaptively to static problems, its accuracy may be improved for dynamic problems. In other words, it is an effective way to improve its accuracy that the viscous-spring dynamic boundary is modified to be a unified boundary for static and dynamic applications.

The accuracy of viscous-spring boundaries for static problems is totally determined by the stiffness K_b. Therefore, it is necessary to modify parameter α according to the fundamental solutions of static problems in half space. Fig. 3 shows the Boussinesq's problem and the Cerruti's problem. In both problems, the ratio of the stress to corresponding displacement on the given section can be written in the form of eq. (1), which is shown in Table 2, where parameter α is the function of cylindrical coordinates r, z or rectangular coordinates x, y, and z.

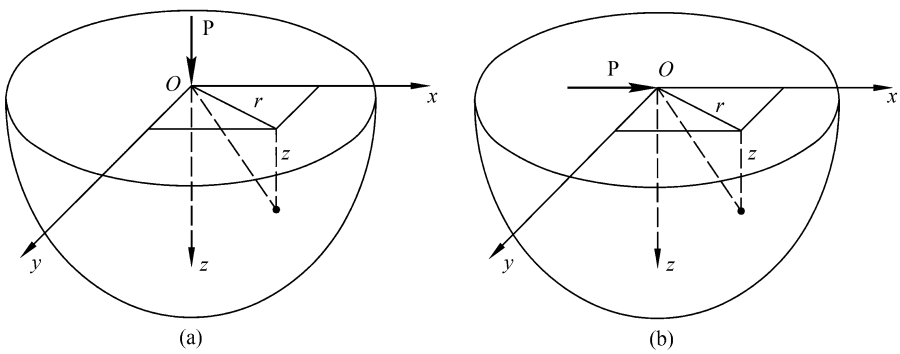

Fig. 3. Half space with concentrate force applied on its free surface. (a) Boussinesq's problem; (b) Cerruti's problem.

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Table 2 Ratio between stress and displacement in Boussinesq's problem and Cerruti's porblem

	Position	Normal direction	Tangent direction
Boussinesq's problem	$z = z_b$	$K_{b1} = -\frac{\sigma_z}{u_z} = \alpha_1 \frac{G}{z_b}$	$K_{b2} = -\frac{\tau_{rz}}{u_r} = \alpha_2 \frac{G}{z_b}$
	$r = r_b$	$K_{b3} = -\frac{\sigma_r}{u_r} = \alpha_3 \frac{G}{r_b}$	$K_{b4} = -\frac{\tau_{rz}}{u_z} = \alpha_4 \frac{G}{r_b}$
Cerruti's problem	$z = z_b$	$K_{b5} = -\frac{\sigma_z}{u_z} = \alpha_5 \frac{G}{z_b}$	$K_{b6} = -\frac{\tau_{xz}}{u_x} = \alpha_6 \frac{G}{z_b}$
	$x = x_b$	$K_{b7} = -\frac{\sigma_x}{u_x} = \alpha_7 \frac{G}{x_b}$	$K_{b8} = -\frac{\tau_{xz}}{u_z} = \alpha_8 \frac{G}{x_b}$

Fig. 4 shows the value of parameter α with $\nu = 0.25$, in which R is the perpendicular distance between the loading point and the given section, d denotes the position of the observation point. For α_1 and α_2, $d = r$; for α_5 and α_6, $d = x$; for α_3, α_4, α_7 and α_8, $d = z$. It should be explained that α_5, α_6, α_7 and α_8 tend to decrease if $y \neq 0$. As shown in Fig. 4, there are singular points and negative range in α_1 and α_2, so that K_{b2} and K_{b3} cannot be simulated by ordinary springs in these regions.

Coupling the above solutions and the viscous-spring dynamic boundary, the value of parameter α is suggested as follows with consideration that the static effect in civil engineering is usually induced by gravity: the normal parameter α of the bottom boundary plane is defined as α_1, which is the same to the normal parameter α of lateral boundary plane because of the comparability of α_7 and α_1; the tangent parameter α of the bottom plane is defined as the tangent parameter α of the 3-D viscous-spring dynamic boundary; according to α_4, the tangent parameter α of the lateral plane is taken 0.5, which equals to the tangent parameter α of the 2-D viscous-spring boundary. All the above modifications of parameter α adopted in eq. (1) are shown in Table 3, in which α^* is expressed as follows:

$$\alpha^* = \frac{6}{2(1-\nu)\left[1+\left(\frac{d}{R}\right)^2\right]^2 + \left[1+\left(\frac{d}{R}\right)^2\right]}, \tag{6}$$

where d is the situation coordinate, $d = r$ for the bottom boundary plane, and $d = z$ for the lateral boundary plane; R is the distance between the loading point and the artificial boundary, for concentrate force, R is defined as the perpendicular distance between the loading point and the artificial boundary, for distributed force, R is the perpendicular distance between the center of distributed force and the boundary, and for the dispersion problems, R is the perpendicular distance between the scattering source and the artificial boundary.

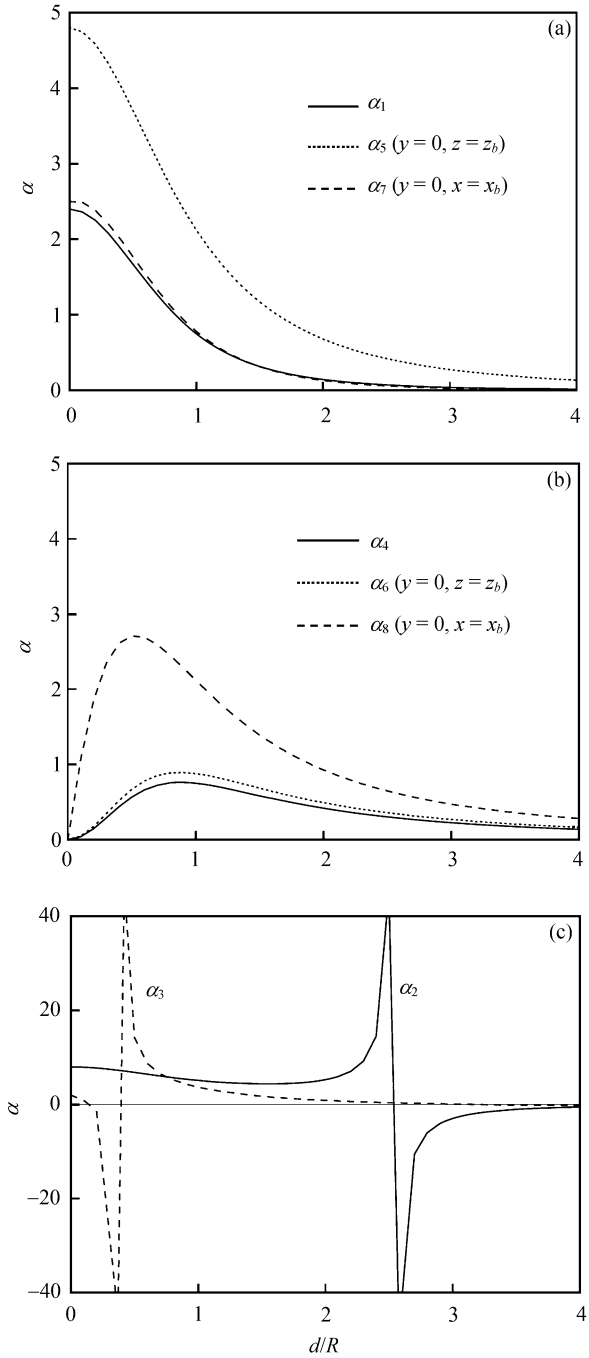

Fig. 4. Parameter α in Table 2.

It is demonstrated by numerical examples that the artificial boundary modified is of high accuracy for both static and dynamic problems in half space, so it can be called the unified viscous-spring artificial boundary for 3-D static and dynamic applications.

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Table 3 Parameter α in the 3-D unified viscous-spring artificial boundary

Position	Direction	α
Bottom plane	Normal	α^*
	Tangent	2.0
Lateral plane	Normal	α^*
	Tangent	0.5

3 Numerical examples

3.1 Problems of half space with a normal concentrate force on its surface

The model is shown in Fig. 3(a), $\rho = 1$, $c_s = 4$ and $\nu = 0.25$. $P = 1$ is used in Boussinesq's problem, whose analytical solution was given by J. Boussinesq[14]. Force $P(t)$ is taken δ function as eq. (5) for Lamb's problem, whose analytical solution can be obtained by the integration of the fundamental solution[15]. The computational range is $-0.5 \leqslant x, y \leqslant 0.5$, $0 \leqslant z \leqslant 1.0$, the finite element size is $\Delta x = \Delta y = \Delta z = 0.05$, and the unified viscous-spring boundary, the 3-D viscous-spring dynamic boundary, the viscous boundary and the roller boundary are applied in this example, R is taken as the perpendicular distance between the loading point and the boundary plane.

The normal displacement time history of the observation points is shown in Fig. 5 with the distance r from the loading points, and the origin (0, 0, 0) and the point A (0.2, 0.2, 0) are considered as loading points respectively. 1 and 2 in Fig. 5(c) denote the observation points (0.2, 0.4, 0) and (0.2, 0, 0) correspondingly. According to Fig. 5, the viscous boundary results in floating instability. The 3-D viscous-spring boundary is stable, but its stiffness is on the high side. The unified viscous-spring boundary has good stability and its accuracy is superior to the other two boundaries. It can also be seen from Fig. 5(c) that the analysis result based on the unified boundary is governed by the distance r between the observation point and the loading point, and whether the observation point is close to the boundary or not has little influence on the result, but for viscous or viscous-spring dynamic boundaries there is an obvious difference. Thus, it is further verified that the unified viscous-spring boundary has better accuracy.

The displacement solution of Boussinesq's problem with force P loading at the origin is shown in Fig. 6. According to Fig. 6, the accuracy of the unified viscous-spring boundary is apparently higher than that of the other two artificial boundaries for the displacement component u_z, and there is no obvious decrease of accuracy even if on the boundary plane $z = 1.0$. It must be noticed that the relation of σ_r and u_r cannot be simulated by ordinary springs when $0.15 \leqslant z/R \leqslant 0.36$, but the accuracy of the unified boundary is still very high.

3.2 Problems of half space with a tangent concentrate force on its surface

As shown in Fig. 3(b), the computational region is taken $-1 \leqslant x, y \leqslant 1$, $0 \leqslant z \leqslant 1$, and the concentrate force in x direction is loading at the origin. The parameter R is

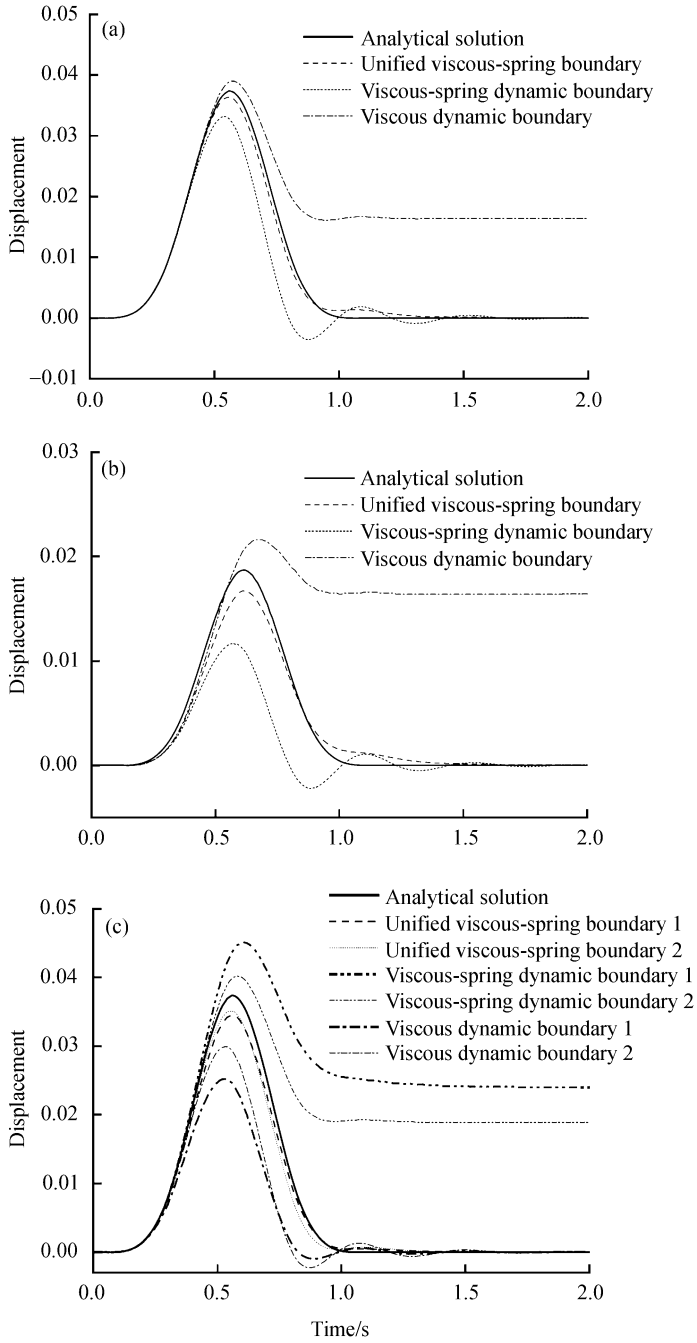

Fig. 5. Displacement of the observation point in z direction in Lamb's problem (vertical force). (a) $r = 0.2$ (loading at the origin); (b) $r = 0.4$ (loading at the origin); (c) $r = 0.2$ (loading at point A).

defined as 1.0 on every boundary and the other parameters are the same as sec. 3.1. The static and dynamic results are shown in Fig. 7 and Fig. 8 separately, in which the ana-

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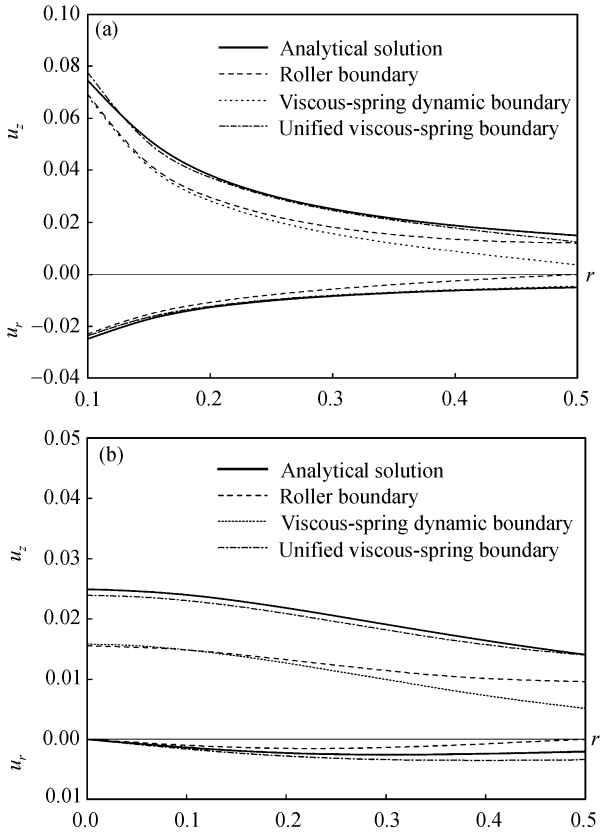

Fig. 6. Displacement solution of Boussinesq's problem (loading at the origin). (a) Plane z = 0.0; (b) plane z = 0.5.

lytical solution of static problem is given by V. Cerruti[14] and that of dynamic problem is the finite element solution of extended boundary. According to Figs.7 and 8, for the dynamic problem under the horizontal force, the 3-D unified viscous-spring boundary is still superior to the other two boundaries. For the static problem, the accuracy of horizontal displacement is relatively high but that in z direction is relatively low. Under the consideration that the static effect is usually induced by the vertical gravity in civil engineering, this disadvantage has little influence on applications of the 3-D unified viscous-spring boundary.

3.3 Problems of the layered half space with normal distributed force on its surface

As shown in Fig. 9, half space $z \geqslant 0$ is constituted by two layers. When $0 \leqslant z \leqslant 0.5$, $\rho = 1$, $c_s = 2$, $\nu = 0.30$; and when $z > 0.5$, $\rho = 1$, $c_s = 4$, $\nu = 0.25$. The applied region of the normal distributed force is $-0.5 \leqslant x, y \leqslant 0.5$, the force $P(t)$ is also taken as δ function given in eq. (5). Other parameters are the same as sec. 3.2. The displacement time histories of two observation points are given in Fig. 10, in which the analytical solution is the finite element solution with extended boundary. According to Fig. 10, the 3-D unified

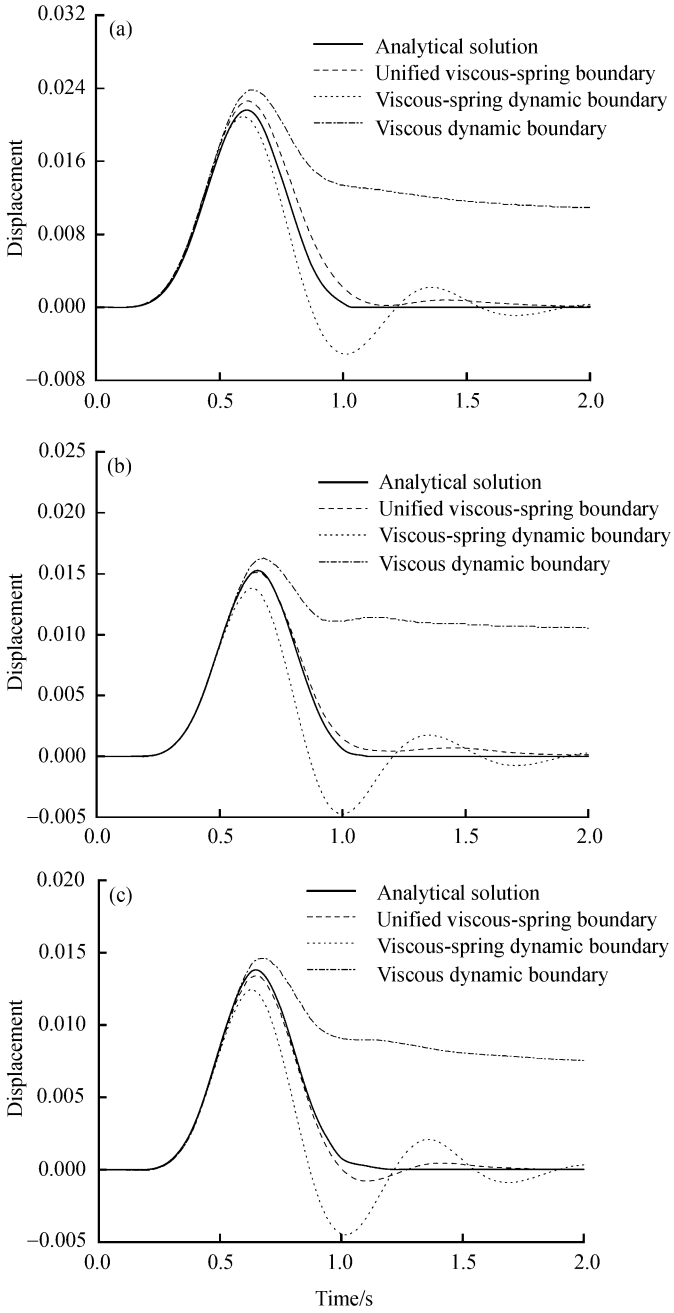

Fig. 7. Displacement time history of observation points in x direction. (a) Point (0.5, 0, 0); (b) Point (0, 0.5, 0); (c) Point (0, 0, 0.5).

viscous-spring boundary can be applied in the dynamic analysis of the layered half space with high accuracy and good stability, which is superior to the other two artificial boundaries.

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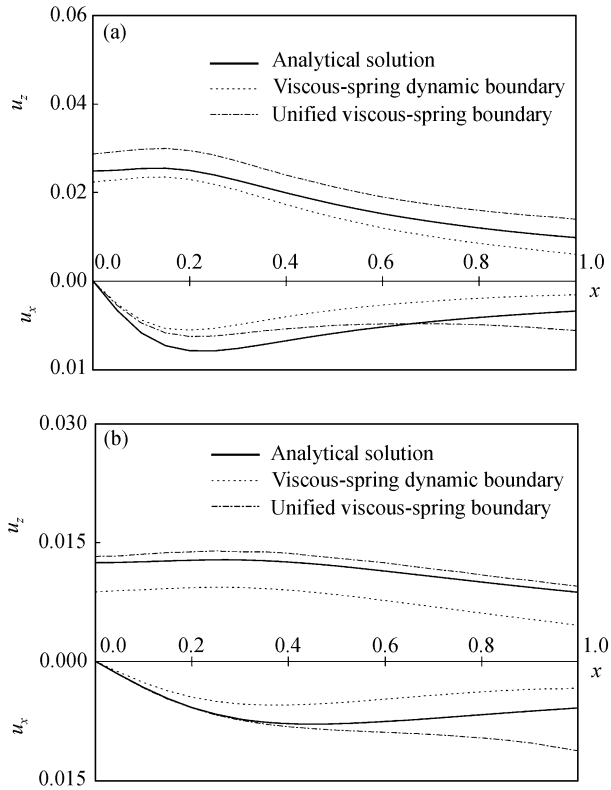

Fig. 8. Displacement solution of Cerruti's problem. (a) $y = 0$, $z = 0.25$; (b) $y = 0$, $z = 0.50$.

3.4 Dispersion problems

Considering half space $z \geqslant 0$, $\rho = 2$ g/cm^3, $c_s = 100$ m/s, $\nu = 0.25$, in which there is a cavity with $-6 \leqslant x,y \leqslant 6$ m, $0 \leqslant z \leqslant 6$ m. The analysis model is depicted in Fig. 11 with symmetry. The whole computational region is taken $0 \leqslant x,y,z \leqslant 20$ m, which is discretized by cubic elements with the size $\Delta x = \Delta y = \Delta z = 2$ m.

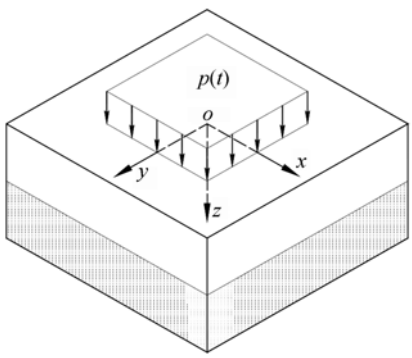

Fig. 9. Analysis model of the layered half space.

The unified viscous-spring boundary, the 3-D viscous-spring dynamic boundary and the viscous boundary are applied, and R is also taken as the perpendicular distance between the origin and the boundary plane. A vertical incident pulse wave $\delta(t)$ with the expression (5) ($T = 0.5$ s) is taken as the excitation, and the wave input method developed in ref. [4] is applied. Fig. 12 shows the displacement histories of observation points A and B, according to which the 3-D unified viscous-spring

boundary is of high accuracy and good stability for dispersion problems.

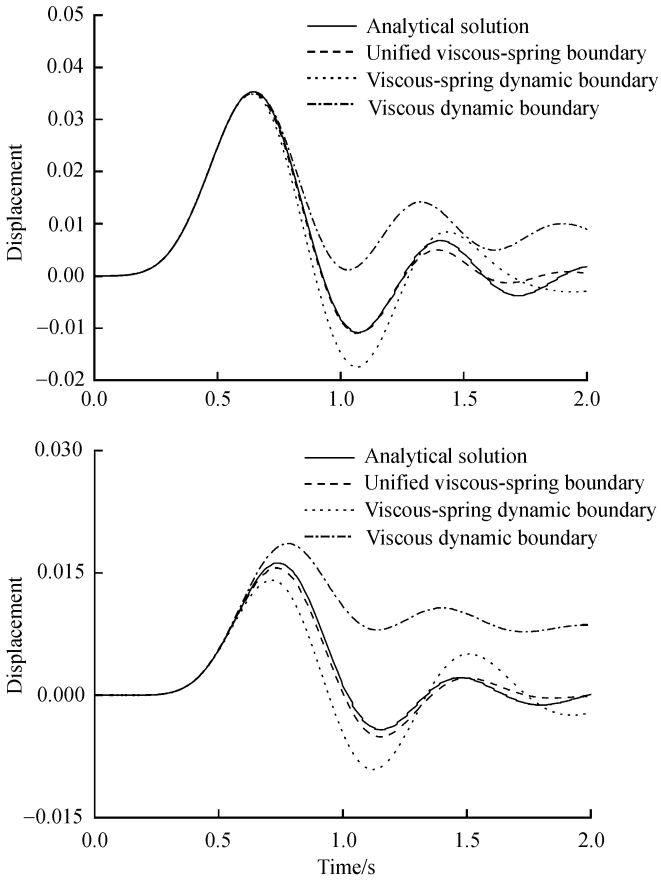

Fig. 10. Problem of the layered half space. (a) Displacement of point (0.5, 0, 0); (b) displacement of point (0.5, 0, 0.5).

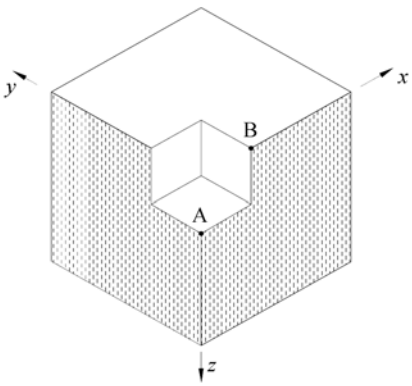

Fig. 11. Analytical model of the dispersion problem.

4 Conclusion

A method to develop a unified viscous-spring boundary for static and dynamic problems is provided in this paper. On the basis of Boussinesq's problem and Cerruti's problem in half space, the 3-D viscous-spring uniform boundary is proposed. The new artificial boundary has the clear definition and simple formulation, and it can be implemented easily in the finite element method. It is verified by numerical examples that for dynamic problems, the unified viscous-spring boundary is superior to the viscous or viscous-

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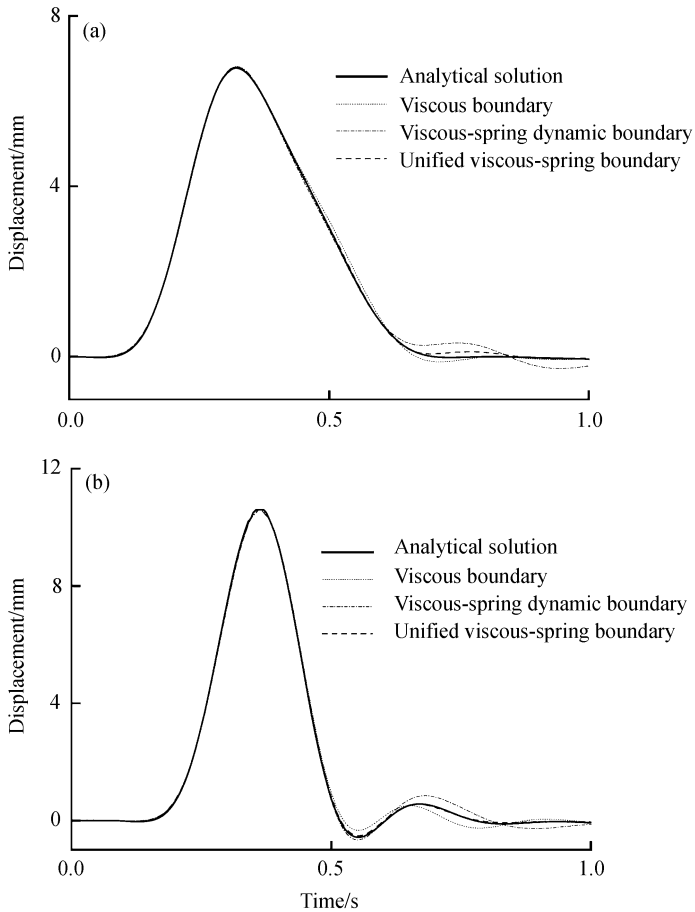

Fig. 12. Displacement histories of observation points in x direction. (a) Point A; (b) Point B.

spring dynamic boundary, which has high accuracy and good stability; for static problems, the unified boundary is still in good accuracy and better than the roller boundary commonly used at present. The proposed unified boundary breaks through the limit of dynamic artificial boundaries that the static and the dynamic effect cannot be considered at the same time, so it can simulate the soil-structure interaction under the influence of static force in civil engineering more reasonably and conveniently.

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References

1. Lysmer, J., Kulemeyer, R. L., Finite dynamic model for infinite media, Journal of Engineering Mechanics, ASCE, 1969, 95: 759－877.
2. Deeks, A. J., Randolph, M. F., Axisymmetric time-domain transmitting boundaries, Journal of Engineering Mechanics, ASCE, 1994, 120(1): 25－42.
3. Liu Jingbo, Lu Yandong, A direct method for analysis of dynamic soil-structure interaction based on interface idea, in Dynamic Soil-Structure Interaction (eds. Zhang, C. H., Wolf, J. P.), Beijing: International Academia

Publishers, 1997, 258－273.
4. Liu Jingbo, Lu Yandong, A direct method for analysis of dynamic soil-structure interaction, China Civil Engineering Journal (in Chinese), 1998, 31(3): 55－64.
5. Wang Zhenyu, Computational theory of dynamic response of large structure-soil systems and its application, Ph. D Dissertation, Tsinghua University, Beijing, 2002.
6. Liao Zhenpeng, Introduction to Wave Motion Theories in Engineering (in Chinese), Beijing: Science Press, 2002.
7. People's Republic of China. GB50267-97, Seismic Design Code of Nuclear Power Plants (in Chinese), Beijing: China Planning Press, 1998.
8. Liu Jingbo, Wang Zhenyu, Zhang Kefeng et al., 3D finite element analysis of large dynamic machine foundation considering soil-structure interaction, Engineering Mechanics (in Chinese), 2002, 19(3): 34－38.
9. Wang Zhenyu, Liu Jingbo, Pei Yuxiao et al., Theoretical and experimental study on micro-vibration reduction of large dynamic machine foundation, Chinese Journal of Geotechnical Engineering (in Chinese), 2002, 24(3): 363－366.
10. Xiong Hui, Zou Yinsheng, Xu Zhenyu, The simulation of interactive effect on pile-soil-structure by whole dynamic finite element method in layered soil field (in Chinese), China Civil Engineering Journal, 2004, 37(9): 55－61.
11. Kellezi, L., Local transmitting boundaries for transient elastic analysis, Soil Dynamics and Earthquake Engineering, 2000, (19): 533－547.
12. Lysmer, J., Analysis method in soil mechanics, in Translations on Earthquake Engineering and Soil Dynamic Problems (in Chinese), Beijing: Seismic Press, 1985.
13. Wolf, J. P., Song, C., Finite-Element Modeling of Unbounded Media, New York: Wiley, 1996.
14. Xu Zhilun, Elasticity Mechanics (in Chinese), Beijing: Higher Education Press, 1990.
15. Pekeris, C. L., The seismic surface pulse, Proceedings of the National Academy of Science, 1955, 41: 469－480.

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